

The Immortality Key as an Aletheia Case: Evidence Map, Countercase, and Research Agenda

Executive assessment

Brian Muraresku's *The Immortality Key* is unusually well suited to Aletheia because the controversy is not really one proposition. It is a **chain of increasingly ambitious propositions**, and the evidence becomes much thinner as one moves from "ancient people deliberately used psychoactive substances" to "the Eleusinian kykeon was psychedelic" to "some early Christians used a psychedelic Eucharist" to "Christianity inherited a continuous pharmacological sacrament from the pagan Mysteries." Muraresku's own public description spans all of those levels: ancient Greek drug use, a visionary Eleusinian drink, Dionysian wine, and a possible psychedelic Eucharist inherited by early Christians. His website also acknowledges that the "smoking gun remains elusive." ¹

My first-pass assessment is:

Atomic proposition	Current assessment
Ancient societies deliberately used potent psychoactive plants in ritual/religious contexts	Strongly established
Such practices existed in the ancient Mediterranean and Near East	Strongly established
Greeks and Romans had the technical knowledge to make pharmacologically active botanical preparations, including wine-based preparations	Well supported
Ergot was present and apparently consumed in a Demeter/Kore-related ritual context at Mas Castellar de Pontós in Iberia	Significant evidence, but the original finding deserves modern reanalysis
A preparation technically accessible to ancient people can convert some toxic <i>Claviceps purpurea</i> ergopeptides into LSA/iso-LSA	Experimentally demonstrated as of 2026
The actual Eleusinian kykeon contained deliberately prepared ergot	Plausible, substantially more interesting than a decade ago, but unconfirmed
Dionysian ritual wine was characteristically psychedelic beyond ethanol	Possible but unconfirmed
Some early Christian groups put psychoactive substances into Eucharistic wine	Plausible as a research hypothesis, presently weakly evidenced

Atomic proposition	Current assessment
The earliest Eucharist or the Last Supper itself was psychedelic	Very weakly evidenced
Christianity inherited its Eucharist from an Eleusinian/Dionysian psychedelic sacramental tradition	Not established; the transmission bridge is the weakest part of the case
A continuous “religion with no name” runs from prehistory through Eleusis into Christianity	Too composite to score as one historical claim; it should be decomposed into separate transmission claims

The important update since *The Immortality Key* appeared in 2020 is **not** “Muraresku has been proved right.” Rather, several objections that once made parts of the hypothesis look intrinsically unlikely have weakened. Direct chemical evidence now makes deliberate ancient psychoactive consumption difficult to dismiss as a category; a 2026 experiment makes a less-toxic ergot preparation chemically feasible; and archaeochemical work continues to reveal sophisticated compound preparations. But none of those discoveries supplies the missing direct evidence that a vessel from Eleusis contained ergot alkaloids, or that a securely identified early Christian Eucharistic vessel contained a hallucinogen. ²

That distinction is exactly where Aletheia can improve on much of the public debate. Your site says it decomposes controversial hypotheses into atomic claims, keeps “source says” separate from inference, prevents lower-rung evidence from automatically climbing the claim ladder, and asks which experiment could actually resolve the dispute. *The Immortality Key* is almost a textbook stress test for that architecture.

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What Muraresku's case actually consists of

The thesis is better represented as a ladder than as a yes/no question

Muraresku's strongest public rhetoric sometimes sounds sweeping. His own site asks whether ancient Greeks “used drugs to find God,” whether the earliest Christians “inherit[ed] the same, secret tradition,” and whether knowledge of visionary plants, herbs, and fungi was transmitted from the Stone Age onward. It also describes the Eleusinian drink as a “holy beer” that produced heavenly visions and Dionysian wine as a sacrament by which devotees became one with the god. ¹

In a serious scholarly discussion at Harvard Divinity School in 2021, however, Muraresku articulated a substantially narrower position. When Charles Stang asked about the Last Supper itself, Muraresku said he did not “hang [his] hat” on Jesus administering a psychedelic sacrament. He instead called it more promising to investigate whether **some early Christians** may have used one. He explicitly acknowledged that whether Dionysian wine was psychedelic is “a big if,” and then acknowledged a second unresolved step: even if Dionysian wine was psychedelic, did the drugs actually pass into Christianity? ⁴

Stang's response gives Aletheia an excellent decomposition rule. He pointed out that at least three questions had been blended together:

What did Jesus do at the Last Supper?
 What did the earliest Christian communities do with the Eucharist?
 How did the Gospel of John interpret Jesus, wine, and Eucharistic symbolism?

Those propositions can have different answers. A Dionysian literary motif in John, for example, would not establish the recipe of the historical Last Supper. ⁴

The same decomposition is needed on the Greek side:

Were psychoactive preparations known?
 Were they used religiously?
 Were they used in Demeter/Kore cults?
 Was ergot specifically used?
 Was it used at Eleusis?
 Was it the secret ingredient of the canonical kykeon?
 Did it account for the extraordinary experiences reported by initiates?

Evidence for an earlier question alters the prior probability of later ones, but it does not prove them.

An Aletheia-ready claim graph

I would start the case with something like the following graph.

ID	Atomic claim	Best evidence for it	Principal objection / missing edge	My classification
C1	Ancient humans deliberately consumed psychoactive plants in ritual contexts	Datura at Pinwheel Cave; cannabis on a Judahite altar; drug alkaloids in Bronze Age Menorcan hair; psychotropic botanical mixture in a Ptolemaic Bes vessel. ⁵	None to the broad proposition; individual sites still differ in function and certainty.	Established
C2	Mediterranean societies possessed sophisticated pharmacological preparation technology	Dioscoridean drug-wine recipes discussed by Muraresku; Roman Villa Vesuvio pharmacological assemblage; current residue evidence. ⁶	Medical/ pharmacological capability does not imply use in a particular mystery cult.	Well supported

ID	Atomic claim	Best evidence for it	Principal objection / missing edge	My classification
C3	The Eleusinian kykeon was textually associated with barley, water, and pennyroyal/ mint	Ancient textual tradition summarized in current Eleusis research. ⁷	The texts do not mention ergot or alkaline detoxification.	Well established, within textual limits
C4	Ergot was deliberately consumed in a Demeter/Kore-linked context at Mas Castellar	2002 Juan-Tresserras reports, summarized by Antonopoulos et al., find ergot in a ceremonial vessel and human dental calculus in a sanctuary of the "two Eleusinian Goddesses." ⁷	Not Eleusis; old analytical work needs independent modern replication; presence does not determine dose or purpose.	Substantial but not definitive
C5	Ancient-compatible alkaline processing can turn ergot ergopeptides into simpler psychoactive lysergamides	Antonopoulos et al. 2026 experimentally produced LSA and iso-LSA from <i>C. purpurea</i> using alkaline treatment. ⁸	No evidence yet that Eleusinian personnel actually used this protocol.	Experimentally supported feasibility
C6	The resulting preparation could have been psychoactive	LSA/iso-LSA have central serotonergic/ psychoactive activity; literature records hallucinations and other psychotropic effects. ⁹	LSA is much less straightforward pharmacologically than LSD; nausea, autonomic effects, anxiety and other adverse effects occur. ¹⁰	Supported, but "psychoactive" is safer than "classic psychedelic"
C7	The Eleusinian kykeon actually contained prepared ergot	C3+C4+C5 create a coherent circumstantial case. ¹¹	No direct ergot residue from Eleusinian vessels; no textual evidence for the added alkaline processing step. Antonopoulos et al. themselves call for Eleusinian residue analysis. ¹²	Live, plausible, unconfirmed

ID	Atomic claim	Best evidence for it	Principal objection / missing edge	My classification
C8	Dionysian cult wine was systematically psychoactive beyond ethanol	Ancient drug-wine technology plus Dionysian associations with altered states; Muraresku marshals textual parallels. ⁴	There is no comparable chemical “Dionysian ritual wine” smoking gun; Muraresku himself calls the psychedelic premise “a big if.” ⁴	Unresolved
C9	The Gospel of John deliberately engages Dionysian imagery	Dionysian readings of Cana are genuine scholarly possibilities, not an invention of Muraresku; scholarship discusses Dionysus as one possible religio-historical background to the wine sign. ¹³	Other interpretive backgrounds exist, including Jewish/messianic abundance symbolism; literary allusion does not identify a chemical recipe. ¹⁴	Plausible literary claim; significance contested
C10	Some early Christians used psychoactive Eucharistic mixtures	Ancient drug technology was available; Christians inhabited the same Mediterranean environment; Stang considers such use plausible enough to test. ¹⁵	No securely contextualized early-Christian vessel has yet supplied direct evidence; Muraresku admits his evidence relies heavily on John and Italy. ¹⁵	Possible, currently low direct support
C11	The original Eucharist/Last Supper was psychedelic	Mostly extrapolation from C8–C10. Muraresku explicitly treats it more cautiously in scholarly discussion. ⁴	No archaeological evidence for the original Eucharist; the Jewish context and earliest Christian textual evidence must be independently explained. Muraresku himself states the archaeological absence. ¹⁶	Very weakly supported

ID	Atomic claim	Best evidence for it	Principal objection / missing edge	My classification
C12	Christianity inherited a pharmacological sacrament through historical continuity with Greek mysteries	Possible cultural contact, literary overlap and common beverage symbolism.	No demonstrated transmission chain from Eleusinian priesthood or Dionysian groups to a Christian Eucharistic community; each inferred bridge depends on earlier uncertain claims. ¹⁷	Not established

The critical Aletheia behavior here is that **C1 does not automatically support C7 very strongly**, and **C7 does not automatically support C10**. The graph should encode *distance*. A chemically proven Bronze Age hallucinogen in Menorca is excellent evidence against “ancient Europeans did not ingest psychoactive plants,” but very poor direct evidence that a Christian in Corinth put one into a Eucharistic cup a millennium later. ¹⁸

The source corpus Aletheia should ingest

The best corpus is deliberately heterogeneous. It should contain Muraresku and his intellectual predecessors, the underlying excavation and chemistry, skeptical classicists and historians of religion, and the newer findings that change background probabilities. It should **not** treat all of those source types as equivalent.

Foundational sources for the positive case

Source	Why it belongs in the corpus	What Aletheia should extract
R. Gordon Wasson, Albert Hofmann & Carl A. P. Ruck, <i>The Road to Eleusis</i> (1978; expanded anniversary edition 2008)	Foundational formulation of the ergot/kykeon theory subsequently developed by Muraresku. It is explicitly identified by the 2026 experimental paper as the origin of the hypothesis. ⁷	Every distinct proposed ingredient; arguments from barley; Hofmann's pharmacological reasoning; assumptions about ancient ergot chemistry; distinction between possibility and asserted history.
Peter Webster, Carl Ruck & Daniel Perrine, “Mixing the Kykeon” (2000)	Pre-2026 proposal that alkaline processing could alter ergot chemistry; directly cited as antecedent to the new experiment. ⁷	Exact proposed preparation steps and which are textually/ archaeologically attested versus reconstructed.

Source	Why it belongs in the corpus	What Aletheia should extract
Brian Muraresku, <i>The Immortality Key</i> (2020)	The claim bundle being tested. Muraresku links Eleusis, Dionysian wine, Greco-Roman pharmacology, and possible early-Christian sacramental use. ¹	Sentence-level claims tagged <code>observed</code> , <code>source_report</code> , <code>inference</code> , <code>speculation</code> , and especially every appearance of “smoking gun,” “proof,” “evidence,” “could,” “may,” and “if.”
Muraresku–Stang Harvard Divinity School discussion (2021)	Arguably more useful than many reviews because Muraresku explicitly states which claims he does and does not wish to defend strongly. ¹⁹	Differences between book rhetoric and later qualified formulation; explicit concessions and proposed tests.
Jordi Juan-Tresserras, Mas Castellar de Pontós residue/ microremain chapters (2002)	The archaeological hinge in the Eleusinian case: ergot associated with a ritual vessel and dental calculus at a Demeter/Kore-related sanctuary. Antonopoulos et al. identify the original chapters precisely. ⁷	Original laboratory technique, sample IDs, stratigraphy, contamination controls, what was <i>actually</i> identified, and whether beer/yeast and ergot came from exactly the same sample/context.
Marina Ciaraldi, <i>People and Plants in Ancient Pompeii</i> (2007), especially Villa Vesuvio	One of Muraresku's main “proof of concept” cases for Roman pharmacological sophistication. A scholarly review reports that more than half of the waterlogged material in the Scafati dolium was classified as pharmacological by comparison with Dioscorides and Pliny. ²⁰	Vessel-level context, plant identifications, whether materials were mixed simultaneously, whether wine itself was detected, and whether “pharmacy/drug farm” is stronger than the data warrant.
Dioscorides, <i>De materia medica</i>	Muraresku uses it to establish that first-century pharmacology included botanicals such as henbane/ mandrake/nightshade in wine preparations. ⁴	Exact Greek, recipe, dosage, indication, expected effect and toxicity. Critically: therapeutic drug-wine recipes should not be silently reclassified as mystery-cult sacraments.

One especially valuable Aletheia exercise would be to reconstruct Muraresku's argument **without his prose**, using only the original sources behind each footnote. The degree to which the same thesis survives that transformation is itself diagnostic.

The most important new scientific evidence

Antonopoulos et al., “Investigating the psychedelic hypothesis of kykeon, the sacred elixir of the Eleusinian Mysteries,” *Scientific Reports* (2026) is now indispensable. It is the strongest post-book development specifically addressing Eleusis rather than ancient psychedelics generally. The researchers performed 48 experiments and found that under one tested condition—5% w/v ergot, an initial lye pH of 12.5, and at least 120 minutes of heating—the monitored toxic ergopeptides were completely hydrolyzed

while LSA and iso-LSA appeared. They quantified roughly 0.54 mg LSA and 0.48 mg iso-LSA per gram of ergot under their highlighted condition. ²¹

This paper should be marked **strong evidence for chemical feasibility, not direct evidence for ancient practice**. Its authors explicitly describe the goal as demonstrating feasibility in a historical-technological context and conclude by calling for advanced organic-residue analysis of artifacts from Eleusis. ²²

That distinction matters because the experimental recipe adds an unattested step. The ancient kykeon tradition cited by the authors gives water, barley and pennyroyal/mint; the experimentally successful preparation relies on highly alkaline treatment. Ancient knowledge of alkalis establishes technological availability, but there is presently no direct evidence in the paper that the Eleusinian recipe actually included an ash/lye process. ²³

Aletheia should therefore represent the inference as:

Ancient people could make alkaline solutions + alkaline treatment can convert modern ergot chemistry → Eleusinian personnel could theoretically have done so

not:

Eleusinian personnel did so.

There are further questions the paper does not settle. Its ergot included modern material from collections and modern field/grain sources, including material obtained in Alberta; the authors provide evolutionary arguments for continuity of major ergot chemistry across historical time, but this remains different from analyzing ancient Greek ergot. ²²

Likewise, the paper's chemical safety argument should not be promoted to "the resulting beverage is demonstrated safe for humans." The study shows disappearance of the monitored major ergopeptides and reports no abundant unexpected known toxicologically relevant ergot products in its analytical screens; it is **not a human safety trial**. ²⁴

Finally, "LSA was produced" should not silently become "an LSD-like mystical experience would predictably result." A recent systematic review characterizes LSA as psychoactive and reports euphoria and hallucinations, but also nausea, anxiety and potentially serious adverse outcomes; older forensic human data associate LSA exposure with prominent vegetative adverse effects. ¹⁰

That pharmacological gap is a particularly good Aletheia subcase.

Sources for the skeptical and alternative case

The strongest scholarly counterargument published since the book is **Sharday Mosurinjohn and Richard Ascough, "Psychedelics, Eleusis, and the Invention of Religious Experience," first published online in 2025 and appearing in *Psychedelic Medicine***. Their central methodological criticism is almost tailor-made for Aletheia: they argue that the psychedelic-Eleusis literature repeatedly presents a possibility, offers mild

circumstantial support, rhetorically hardens the interpretation into a fact, and then uses that newly hardened “fact” as the premise for the next conjecture. ²⁵

Their article is useful, but Aletheia should not simply substitute their verdict for Muraresku's. Their abstract goes beyond evidential criticism into a critique of contemporary psychedelic legitimization and calls the history “pseudo-history” and a form of religious fundamentalism. Those claims concern intellectual genealogy and modern ideology rather than the archaeological truth of the kykeon recipe. They should therefore be tagged separately. ²⁵

There is also a timing issue worth explicitly representing. Mosurinjohn and Ascough's paper first appeared July 25, 2025, before the full 2026 *Scientific Reports* alkaline-hydrolysis paper. The 2026 chemistry weakens one possible skeptical objection—“a workable ergot preparation is chemically implausible”—but it does **not** answer their central historical objection about inference from possibility to use. ²⁶

Charles Stang's Harvard exchange with Muraresku is a more finely grained counter-source because Stang does not dismiss the project. He accepts that ancient psychoactive biotechnology existed and says he would not be surprised if some Christian group eventually turned out to have used it. But he emphasizes that finding one such Eucharistic vessel would not establish the practice as the normative Eucharist. That is precisely the distinction the case needs. ¹⁵

Kevin Clinton, a leading Eleusis specialist, is important to include directly. In a 2025 investigative article, he is quoted saying that no evidence of drug use has been found at Eleusis and strongly criticizing the methodology of beginning with the psychedelic hypothesis. ²⁷

Patrick McGovern, a major figure in archaeological chemistry who advised Muraresku, is another necessary source—not because expert authority decides the issue, but because he disputes Muraresku's interpretation of material he helped evaluate. McGovern told *Reason* that Muraresku “misconstrues and overinterprets” limited and ambiguous archaeological/archaeobotanical data and characterized scientific support for a psychedelic-Christianity connection as extremely weak. ²⁷

That interview is secondary journalism, so Aletheia should classify it as **expert testimony about interpretation**, not archaeological evidence. Where McGovern, Clinton, Carl Ruck, or David Hillman make concrete claims about a dataset, the next step should be to attach their claim to the original archaeological report.

The corpus should also include mainstream treatments that explain Eleusis without requiring a drug: **George Mylonas's *Eleusis and the Eleusinian Mysteries*; Walter Burkert's work on Greek religion and mystery cults; Helene Foley's *Homeric Hymn to Demeter*; Yulia Ustinova's *Divine Mania*; and recent work on embodied, collective and altered religious experience.** These are not all “anti-psychedelic” sources; their value is that they supply alternative causal models—fasting, anticipation, secrecy, procession, darkness/light, collective ritual, performance, sensory manipulation and culturally structured expectation—which prevent a false dichotomy between “ergot” and “nothing happened.” The 2025 Cambridge treatment of mystery cults, for example, foregrounds embodiment, group experience and ritual as mechanisms of religious intensity. ²⁸

The Christian side needs a stronger independent corpus

This is where I think *The Immortality Key* case most needs improvement.

Muraresku's evidence here should not be evaluated principally by asking whether ancient pagans had drugged wine. It needs its own corpus on **first-century Judaism, earliest Christian meals, Eucharistic development, Johannine composition, Dionysian reception and archaeological identification of Christian drinking vessels.**

There really is scholarship treating the Cana miracle in relation to Dionysus. Harold Attridge notes that interpretations of Cana have sometimes depended on a Dionysian religio-historical background, while other scholarship treats abundant wine as messianic or eschatological symbolism; other specialist literature explicitly analyzes John 2 as responding to Dionysian motifs. ²⁹

That means a fair Aletheia entry should **not** say:

"John/Dionysus parallel = fringe claim."

But it should equally reject:

"John/Dionysus parallel = psychedelic Eucharist."

At most, a Dionysian literary relationship supports a claim about the evangelist's cultural-symbolic environment. To get to psychedelic Christianity requires at least two additional premises: that the relevant Dionysian wine itself contained a psychoactive drug, and that the relevant Christian community transferred the pharmacological practice rather than merely wine symbolism or literary imagery. Muraresku himself acknowledges both inferential gaps. ⁴

The Jewish side deserves special attention. Stang challenged Muraresku explicitly on whether the argument reads Jesus too much through later Greek contexts rather than Jesus and the earliest Christians as Jews. Muraresku agreed this was a major area for further investigation and acknowledged that his proof-of-concept case leaned heavily on the Gospel of John and Italian evidence. ¹⁵

For Aletheia, therefore, **1 Corinthians 10–11, the Synoptic meal traditions, Jewish wine/meal practice, and the chronology of John's Gospel should be sibling evidence streams, not footnotes to the Dionysian stream.**

What the post-2020 discoveries really change

The Athanatos summary you supplied is directionally important, but Aletheia can make it much more precise by asking **which claim each discovery updates.**

Pinwheel Cave does not prove Eleusis, but it demolishes a methodological excuse

The November 2020 Pinwheel Cave study in *PNAS* identified chewed *Datura wrightii* quids at a California rock-art site and described the result as unambiguous confirmation of hallucinogen ingestion at a rock-art site. ³⁰

Its relevance to *The Immortality Key* is **methodological and anthropological, not genealogical**. California is not an evidence link between Eleusis and Christianity. The finding does show that archaeologists can sometimes move beyond speculative iconography and actually connect a psychoactive plant, human consumption and ritualized space.

Aletheia should give it a strong positive edge to:

Ancient ritual ingestion of hallucinogens occurred and can leave detectable material traces.

It should give it effectively no direct edge to:

The Eleusinian kykeon contained ergot.

That prevents the common “pile of fascinating psychedelics = proof of my particular psychedelic hypothesis” error.

Bronze Age Menorca is far more relevant geographically

The April 2023 *Scientific Reports* study analyzed Late Bronze Age hair from the burial and cult cave of Es Càrritx on Menorca using UHPLC-HRMS. The researchers detected **ephedrine, atropine and scopolamine** and concluded that the people represented by the hair had consumed multiple alkaloid-bearing plants by the beginning of the first millennium BCE. ³¹

This is stronger for Muraresku's **background case** than Pinwheel because it is Western Mediterranean evidence and directly measures compounds incorporated into human hair, rather than merely finding a useful plant near ritual activity. The study itself calls this direct evidence of consumption of plant drugs. ³¹

What it does **not** tell us is whether the compounds were consumed specifically during a particular ceremony, whether their purpose was visionary rather than therapeutic/social, or whether such practices were culturally connected to later Greek mysteries. Those remain distinct claims.

So I would attach it:

Strongly → C1: Mediterranean people deliberately consumed psychoactive plants.

Moderately → the prior plausibility of specialist pharmacological knowledge.

Very weakly → C7: ergot at Eleusis.

Almost not at all → C10/C11: psychedelic Christianity.

Tel Arad matters more to the Jewish-background question than is usually appreciated

The 2020 Tel Arad study detected cannabis and frankincense residues on altars from a Judahite shrine; the cannabis resin had apparently been mixed with dung in a manner consistent with controlled heating. ³²

That is important because it directly undercuts an overly strong claim that psychoactive ritual technology was culturally alien to the ancient religious world from which Judaism emerged. Muraresku himself highlighted the finding in his Harvard discussion as archaeochemical evidence from an eighth-century BCE Judahite context. ¹⁵

But again, it is a **prior update**, not a missing Eucharistic link. Burning cannabis on a cultic altar in Iron Age Judah is not the same practice as ingesting a botanical Eucharistic drink centuries later. Aletheia should explicitly encode differences in **date, route of administration, cult, substance and ritual function**.

The Egyptian Bes vessel is impressive, but its provenance caveat matters

The November 2024 *Scientific Reports* study subjected residue from a Bes-shaped Egyptian drinking vessel to a multimodal suite including metabolomics, proteomics, genetic techniques and spectroscopy. It reported botanical signals associated with *Peganum harmala*, blue-water-lily-type *Nymphaea*, and *Cleome*, along with fermented fruit-derived material and honey/royal-jelly-type ingredients. The authors interpret the mixture as a complex ritual preparation containing psychotropic and medicinal substances. ³³

For *The Immortality Key*, this is a powerful demonstration that **complex, fermented, botanical ritual concoctions are archaeochemically realistic in the Hellenistic Mediterranean world**. It makes blanket skepticism about “spiked ceremonial drinks” harder to defend. ³³

However, Aletheia should preserve a caveat that popular summaries easily lose. The sampled Tampa Museum object was bought from a Cairo antiquities dealer in 1960. Its supposed Fayum origin and second-century BCE date lacked supporting excavation documentation; the researchers use typological/mold comparison to bolster Egyptian provenance. ³³

That does not invalidate the chemistry. It changes the confidence of claims about **precise archaeological context and ritual use**.

This is exactly why provenance needs its own axis rather than being collapsed into a single evidence-strength score.

The 2026 kykeon experiment is the one update that directly affects Eleusis

The Antonopoulos et al. paper is qualitatively different from the preceding discoveries because it tests a **load-bearing mechanism in the Eleusinian hypothesis itself**. Its experimentally supported proposition is:

Under specific alkaline-processing conditions, *Claviceps purpurea* material can be converted from a profile containing major toxic ergopeptides into one producing measurable LSA and iso-LSA. ²¹

That answers a genuine feasibility question. It makes “ergot necessarily means catastrophic ergotism” too simplistic. The paper’s authors explicitly frame this as an answer to the toxicity objection. ²⁴

But the historical proposition remains:

Did Eleusinian specialists know and use this particular method?

No residue, installation, textual recipe or other evidence currently establishes that. The authors themselves end by pointing to direct Eleusinian residue analysis as the next evidentiary step. ²²

Aletheia should therefore mark the 2026 study as:

High support for mechanism / feasibility.

Low-to-moderate indirect support for historical plausibility.

No direct observation of Eleusinian practice.

That three-part classification is far more informative than either “2026 study proves the kykeon was psychedelic” or “it proves nothing.”

The Mas Castellar wording is an excellent provenance test case

There is a particularly useful source-propagation issue here.

The 2026 peer-reviewed paper describes the Mas Castellar evidence as **fragments of ergot inside a ceremonial vessel and in human dental calculus**, citing Juan-Tresserras’s original 2002 chapters. ⁷

Some downstream popular coverage has converted this into a statement that “hallucinogenic ergot alkaloids” were found on pottery and in dental calculus. ³⁴

Those are not equivalent claims.

For Aletheia I would preserve:

Original-level claim: *Claviceps purpurea* microremains/fragments were identified.

separately from:

Downstream interpretation: psychoactive ergot alkaloids were chemically detected.

Until the original analytical documentation establishes the latter, the interface should flag the stronger wording as **source drift**.

That one example could be a useful showcase of why exact provenance matters.

Assessment of the positive case and the countercase

The strongest part of Muraresku's case is now the middle, not the endpoint

Before the recent archaeochemistry, the hypothesis could be dismissed too easily by stacking assumptions:

ancient ritual drugs were rare → drugged beverages were implausible → ergot would poison everybody → therefore Eleusis could not have used pharmacology.

That chain is no longer defensible in that simple form. Ancient ritual psychoactive use is directly demonstrated in multiple contexts; sophisticated compound preparations are empirically attested; pharmacological wines are textually and archaeobotanically plausible; and the 2026 paper supplies a workable chemical pathway for ergot. ³⁵

So Muraresku and his predecessors deserve more credit on capability and plausibility than some older skeptical rhetoric allows.

The Mas Castellar evidence is particularly suggestive. It is not merely “some ancient people ate ergot.” On the account reproduced in the 2026 study, the material occurs in a sanctuary dedicated to the two Eleusinian Goddesses, in a ceremonial vessel and human dental calculus. That combination—substance, ingestion evidence and cultic context—is much closer to the hypothesis than generic ancient drug use. ⁷

But geographic transplantation is still required. Mas Castellar was in the orbit of Emporion in Iberia, not the sanctuary at Eleusis in Attica. Evidence that an ergot-associated practice could exist in one Demeter/Kore-related setting does not prove it was the secret recipe administered at the mother sanctuary. The correct inference is **analogy and cultural plausibility**, not identity. ⁷

The biggest remaining weakness at Eleusis is no longer “could it work?” but “did they do it?”

This changes the debate substantially.

After 2026, arguing primarily that ergot is too toxic is less compelling because a tested chemical procedure can remove the monitored complex ergopeptides while producing simple lysergamides. ²⁴

The much better skeptical questions are now:

Where is the Eleusinian ergot?

Where is evidence for the alkaline-processing step?

Does the final reconstructed preparation have the pharmacology necessary to account for the reported experience?

Could it be manufactured consistently at ritual scale from variable natural ergot?

Does it fit the archaeological infrastructure and known ritual sequence?

The 2026 experiment used an initial pH of 12.5, heating and a two-hour reaction for the complete-hydrolysis condition highlighted by the authors. Lower-pH/shorter treatments left toxic ergopeptides detectable. ⁸

That makes historical reconstruction much more specific. We are no longer asking merely whether a priestess could throw contaminated barley into water. A particular processing technology is required. Aletheia should therefore create a separate claim:

C5b: The Eleusinian personnel employed a sufficiently alkaline and controlled processing method.

At present, I would mark C5b **unsupported but testable**.

LSA pharmacology deserves more scrutiny than it receives

A tendency in the pro literature is to move from “LSA is a lysergamide related to LSD” to an intuitive picture of an LSD-like Eleusinian mass vision. That is too quick.

The pharmacological literature supports LSA as centrally active and capable of producing hallucinations/ euphoria, but it is substantially less potent than LSD in relevant receptor assays, and the human literature contains prominent nausea, weakness, autonomic effects, anxiety and occasional severe psychiatric or cardiovascular outcomes. ³⁶

Thus four separate claims are needed:

LSA reaches the brain.

LSA is psychoactive.

LSA can produce hallucinations.

An LSA/iso-LSA-rich ergot extract at an achievable dose reliably produces the kind of profound, largely positive, religious experience inferred for Eleusis.

The first three have meaningful support. The fourth is much less settled.

That does not falsify an ergot kykeon. It creates a researchable pharmacology problem rather than allowing “related to LSD” to do argumentative work by association.

The critical case is strongest when it attacks inference, weakest when it attacks possibility

Mosurinjohn and Ascough's core complaint—that circumstantial evidence can be rhetorically promoted into a fact which then becomes the foundation for the next speculative step—is persuasive as a methodological warning. ²⁵

Their broader effort to classify “psychedelic Eleusis” simply as myth is less decisive after the 2026 chemistry, because there is now an experimentally demonstrated mechanism and there remains the Mas Castellar cultic evidence. The hypothesis is not equivalent to a proposition invented without empirical anchors. ¹¹

I would therefore not have Aletheia label Eleusinian ergot **debunked**.

I would label it:

A materially plausible historical hypothesis with significant circumstantial evidence but no direct evidence at the target site.

That wording preserves both the genuine progress of the positive case and the strongest criticism.

The Christian extension is much weaker than the Eleusinian hypothesis

Here I land considerably closer to the skeptics.

Villa Vesuvio can establish Roman pharmacological capability. Dioscorides can establish recipes. A Dionysian reading of John can establish a possible literary/cultural relationship. None of those supplies the missing observation:

A Christian Eucharistic community consumed a psychoactive sacrament.

Muraresku openly calls Villa Vesuvio a “proof of concept”; in the Harvard exchange Stang emphasizes that nothing about such a Roman pharmacy by itself establishes a Christian connection. Muraresku then acknowledges that the link is circumstantial and says future chalice testing is needed. ¹⁹

The Christian case also suffers from **selection effects**. If one searches Greek and Roman material for examples of pharmacological wine, finds them, then places those examples beside Christian wine symbolism, the resulting juxtaposition feels much stronger than it is. To measure the inference properly, Aletheia needs the background rate:

Among Greco-Roman wines generally, how common were potent botanical additives?

and then:

Among explicitly sacred wines?

then:

Among Dionysian wines?

then:

Among early Christian meal vessels?

Without the denominators, “there existed drugged wine near the period when Christianity emerged” is capability evidence, not transmission evidence.

Muraresku's own later qualifications should be part of the case

This is important for fairness.

Aletheia should not build the strongest imaginable version of “Jesus gave everyone psychedelics at the Last Supper” and label that *Muraresku's claim* when he has publicly disavowed hanging his hat on it. In 2021 he instead proposed that **some early Christian communities** may have interpreted their tradition through a pharmacological sacrament. ⁴

At the same time, Stang immediately observed that this was more hedged than portions of the book. ⁴

That creates a useful provenance feature:

Claim as stated in book → claim as stated by author in later discussion → strongest popular paraphrase.

Those should coexist rather than overwrite one another.

Funding and institutional context should be metadata, not an ad hominem filter

The Athanatos connection belongs in Aletheia, but it needs careful handling.

Yale's Peabody Museum describes the **Yale Ancient Pharmacology Program (YAPP)** as a transdisciplinary effort using organic residue analysis to extract phytochemical fingerprints from ancient artifacts and integrate chemistry with archaeological and textual context. Andrew Koh was appointed Museum Scientist at Yale's Peabody Museum in 2022, when YAPP was recently established. ³⁷

Athanatos currently features YAPP as an active supported project and describes its goals as systematic organic residue analysis, reconstruction of ancient wine-based medicinal and ritual substances, and creation of replicable/open phytochemical archives. ³⁸

The public sources I reviewed support a **real Athanatos–YAPP funding/support relationship**, but I did not find an independent public source in this pass that verifies the precise historical formulation in the supplied text that YAPP was “the first grantee in the Athanatos portfolio upon its founding in 2022.” That sentence should therefore be attributed to Athanatos unless organizational records independently confirm it. Yale-linked research has publicly acknowledged Athanatos among YAPP's supporters. ³⁹

Funding should create a **provenance/funding** edge, not a negative **evidence** edge. The relevant question is whether methods, samples, controls and interpretations stand up to independent replication.

The same rule applies to proponents. Carl Ruck's participation in research on the hypothesis he helped originate is a reason to value **independent replication**, not a reason to discard the research.

A source set I would prioritize for Aletheia

Rather than ingesting everything indiscriminately, I would build the first case around roughly these source clusters.

Core hypothesis and primary archaeological record

Must-have first tier

The Immortality Key itself; *The Road to Eleusis*; "Mixing the Kykeon"; Juan-Tresserras's two 2002 Mas Castellar chapters; the excavation publications for Mas Castellar/Emporion; Antonopoulos et al. 2026; Ciaraldi's Villa Vesuvio publication; the relevant passages of Dioscorides; the Homeric Hymn to Demeter plus a major modern commentary; Clement of Alexandria's description of Eleusinian rites; and the archaeological publications for Eleusis itself. The 2026 study gives a useful bibliography back into most of the core Eleusinian material. ⁷

For Mas Castellar in particular, **getting scans of the original 2002 report should be a priority**. Too much of the current debate relies on later summaries of what was found. The exact sample context and analytical method have exceptionally high resolution value.

Counter case and mainstream contextual scholarship

I would include Mosurino & Ascoug; Mylonas; Burkert; Ustinova; Clinton's major Eleusinian work; and recent mystery-cult scholarship. These sources allow a non-pharmacological model of Eleusinian experience to be built rather than merely saying "there is no drug evidence." ²⁸

The *Reason* investigation is worth ingesting as a **map to dissenting experts**, especially Patrick McGovern and Kevin Clinton, but their claims should then be followed into their scholarly publications rather than allowing a magazine interview to function as the terminal authority. ²⁷

Early-Christian transmission corpus

A separate collection should cover:

the Pauline meal/Eucharist material, Synoptic meal traditions, the Gospel of John, Ignatius, Justin, Clement and other second-century Christian evidence; Jewish drinking and ritual-meal practices; Dionysian cult in the eastern Mediterranean; and the specialist scholarship on John and Dionysus.

The reason to split this from the Eleusinian corpus is methodological. Even a decisive positive answer on Eleusis **would not answer the Christian question**.

There is enough genuine Johannine/Dionysian scholarship that the literary connection should be treated seriously. But alternative readings of the Cana abundance motif must appear symmetrically. ²⁹

Archaeochemical comparison corpus

The following should be attached as **comparative evidence**, not as direct evidence for Eleusis:

Robinson et al., 2020, Pinwheel Cave/Datura. Demonstrates direct hallucinogen ingestion associated with a rock-art setting. ³⁰

Arie et al., 2020, Tel Arad cannabis/frankincense. Demonstrates psychoactive cannabis in a Judahite shrine context. ³²

Guerra-Doce et al., 2023, Es Càrritx, Menorca. Direct human hair evidence for ephedrine, atropine and scopolamine in the Late Bronze Age Western Mediterranean. ³¹

2024 Ptolemaic Bes-vessel study. Multimodal evidence for a fermented botanical concoction containing plants identified as psychotropic/medicinal. ³³

YAPP's newer work. Yale now operates an active program specifically designed to combine organic residue chemistry with textual, ethnobotanical and contextual evidence, and its 2025 fieldwork included vessel sampling across Greece, Italy and Turkey. ⁴⁰

This cluster changes the **base-rate model**. It makes deliberate ancient psychoactive pharmacology an empirically grounded phenomenon rather than an exotic prior assumption.

It does not supply the site-specific bridge.

Research agenda and experiments that could actually move the case

This is where I think Aletheia can outperform both the book and most responses to it. The controversy has several experiments with unusually high resolution value.

Test Eleusis directly

The single highest-value intervention is **systematic organic-residue analysis of authenticated vessels from secure Eleusinian contexts**.

The 2026 paper reaches essentially the same endpoint: advanced residue analysis of archaeological material from Eleusis could provide the missing material evidence. ²²

A strong design would not start by selecting one suspicious-looking cup. It would preregister:

- secure Eleusinian ritual contexts;
- candidate vessel classes;
- chronological strata;
- soil/sediment blanks;
- contemporary domestic controls;
- vessels from comparable non-Eleusinian sanctuaries;
- targeted and untargeted chemical analyses.

The analytical target should go beyond intact ergot alkaloids, which may degrade. The project should establish experimentally validated degradation markers for *Claviceps*, investigate fungal DNA/proteomic

markers where preservation permits, examine starch and fermentation signatures, and specify before analysis what combination would constitute a positive identification.

A replicated positive result from several secure Eleusinian vessels would dramatically change C7.

A large, methodologically sensitive negative survey would also be informative—far more so than today's statement that no ergot has yet been detected there—because Antonopoulos et al. note that systematic residue work at Eleusis has not supplied the relevant dataset. ²⁴

Re-test Mas Castellar with twenty-first-century methods

This may be even easier institutionally and has enormous value.

The original Mas Castellar result is currently doing extraordinary argumentative work. It is invoked as the archaeological bridge connecting ergot with worship of the Eleusinian Goddesses. ⁷

The obvious experiment is independent reanalysis of:

- the miniature/ceremonial vessel;
- retained residue or sediment;
- the dental calculus specimen;
- neighboring human remains;
- control vessels from the same site;
- non-cultic contexts from the settlement.

Use LC-HRMS/MS, microscopy, fungal DNA assays where feasible, proteomics/metabolomics and rigorous contamination controls.

Three results would mean very different things:

Ergot independently confirmed, with alkaloid/degradation markers in the vessel and calculus: major strengthening.

Ergot biology confirmed but no evidence tying it to beverage preparation: still interesting, weaker pharmacological conclusion.

Original identification fails replication: major weakening of the modern Eleusinian argument.

For Aletheia's "resolution value" metric, this experiment should be near the top.

Reconstruct the whole proposed kykeon, not just the hydrolysis reaction

Antonopoulos et al. answer a chemistry question very effectively. The next project should answer the **historical recipe** question.

Their key successful condition used 5% w/v ergot, an initial lye pH of 12.5, and two hours of reaction; they then estimate that about a gram of ergot per initiate could provide the relevant LSA/iso-LSA quantities. 41

Now experimentally constrain the preparation using only operations for which there is independent evidence at Eleusis or in closely comparable Greek food/pharmacological practice:

barley form;
water source;
pennyroyal quantity;
ash species and ash-water recipe;
pH attainable without modern measurement;
heating vessel and fuel;
processing duration;
filtration;
neutralization;
storage time;
temperature;
ergot chemotype variation.

The outcome variable should not merely be LSA yield. It should include residual toxic alkaloids, other degradation products, final pH, stability and reproducibility.

The critical historical question is:

Can a realistic Eleusinian operator, using sensory cues rather than a pH meter or chromatography, reproducibly arrive at a preparation with a narrow enough pharmacological profile to serve thousands of initiates?

That is a much stronger test than “can a modern laboratory reproduce the chemistry with materials ancient people could theoretically possess?”

Test whether the pharmacology matches the phenomenology

Aletheia should create a separate pharmacology program around C6.

The current literature establishes LSA as psychoactive but also documents a less clean and less predictable human profile than the intuitive analogy with LSD implies. 10

Research can proceed without immediately administering reconstructed ergot beverages to humans. The useful first stages are:

quantitative receptor profiling of the complete reconstructed extract rather than isolated LSA;
toxicological screening;
pharmacokinetic modeling;
stability and epimerization studies;

comparison across *C. purpurea* chemotypes;
modeling the likely delivered dose after realistic preparation/storage.

Only after safety and ethics were convincingly established would human work even become a separate question.

Importantly, Aletheia should never encode:

hallucinations possible

as equivalent to:

Eleusinian-style mystical transformation explained.

Those are different observables.

Establish the base rate of pharmacological ritual beverages

Muraresku's argument would become much stronger if drugged ritual drinks turned out to be common across a **predefined comparative sample**, rather than repeatedly appearing in spectacular isolated examples.

A high-value survey would blind analysts where possible and compare vessels from:

Demeter/Kore sanctuaries;
Dionysian contexts;
Asclepian/medical contexts;
domestic contexts;
funerary contexts;
other mystery-associated sites.

The result could answer an important Bayesian question:

Is a pharmacologically active botanical mixture unusually associated with mystery/religious vessels compared with ordinary vessels?

This would convert “we keep finding drugs when we look” into a measurable archaeological frequency.

YAPP's explicit aim of combining systematic organic residue analysis with archaeology, textual evidence and ethnobotany is well aligned with exactly this kind of research. ⁴²

Re-open Villa Vesuvio

Villa Vesuvio currently serves as one of the largest inferential jumps in the Christian argument.

Ciaraldi's work supports a pharmacological interpretation of the unusual plant assemblage, but Stang correctly observes that a Roman pharmacy near Pompeii does not establish Christian ritual use. ⁴³

The useful experiment is not debating the word "psychedelic." It is modern chemical reanalysis of surviving material from the assemblage and other associated vessels.

Questions:

Were pharmacologically potent plants actually compounded together?

Was the matrix wine, another fermented liquid, medicine, stored plant stock, or disposal material?

Can individual vessel histories be reconstructed?

Were psychoactive compounds at concentrations consistent with pharmacological use?

That would clarify exactly what Villa Vesuvio can legitimately support.

Search for early-Christian material without assuming the answer

This is probably the decisive long-term test for the book's most controversial extension.

Muraresku and Stang agree that chemically testing appropriate early-Christian vessels is the way forward; Stang says finding one psychoactive Eucharist would be plausible but would still not establish a universal norm. ¹⁵

The sampling strategy needs to avoid circularity. Aletheia should rank contexts by independent probability of being Eucharistic:

- securely identified early-Christian communal-meal assemblages;
- pre-Constantinian church/house-church contexts;
- cups associated with inscriptions/iconography independently indicating Christian ritual;
- comparison with Jewish and ordinary Greco-Roman meal vessels from the same regions and periods.

A positive psychoactive result from a securely identified Christian ritual vessel would be **transformative evidence for C10**.

It still would not prove C11—the Last Supper—or C12—derivation from Eleusis.

Multiple geographically distributed positives, combined with chronology and botanical similarity to pagan ritual mixtures, would be needed before historical inheritance became a serious transmission claim.

Build a transmission test rather than relying on resemblance

The most important conceptual experiment is historical rather than chemical.

For “pagan continuity” to be more than resemblance, the system should look for **transmission markers**:

- same uncommon ingredient combinations;
- same preparation terminology;
- same ritual handling;
- same vessel conventions;
- chronological overlap;
- geographical pathways;
- communities demonstrably in contact;
- texts recognizing or contesting the practice.

Wine + immortality language is low-specificity evidence because both are widespread symbols.

A rare multi-ingredient pharmacological recipe appearing first in a mystery cult and later in a securely Christian setting would be much higher specificity.

This gives Aletheia a general rule:

Similarity evidence should be weighted by how surprising the similarity would be without transmission.

Test competing explanations of Eleusinian altered states

A fair experiment does not merely ask “drug or no drug.”

Recent scholarship emphasizes embodied ritual, collective experience, secrecy and sensory structure as powerful ingredients of mystery-cult experience. ⁴⁴

Aletheia should model at least four competing causal hypotheses:

H-Drug: pharmacology is the principal extraordinary-state generator.

H-Ritual: fasting, procession, expectation, sensory manipulation, collective arousal and ritual drama are sufficient.

H-Mixed: ritual amplifies a modest pharmacological stimulus.

H-Heterogeneous: experiences differed substantially among initiates and across the many centuries of the cult.

The mixed hypothesis is especially important because proving psychoactive chemistry would not prove that chemistry alone “explained Eleusis.”

Likewise, showing that elaborate ritual can generate altered states would not prove that no substance was used.

Recommendations for how Aletheia should represent this case

This case suggests several design principles that go beyond *The Immortality Key*.

First, make the hypothesis graph visibly asymmetric in evidentiary distance. A Menorcan tropane-positive hair sample should appear as very strong evidence for ancient psychoactive ingestion but as faint evidence for Eleusis and almost invisible evidence for Christianity. That visually enforces Aletheia's own rule that lower-rung evidence does not automatically climb to stronger claims. ⁴⁵

Second, separate feasibility from actuality. The 2026 experiment deserves a prominent green edge to “an ancient-compatible ergot detoxification/conversion process is chemically feasible,” but only a dotted edge to “the Eleusinians used that process.” ²²

Third, distinguish compound identification from interpretation. “Atropine detected in hair” is an observation. “A shaman administered henbane in an initiation ceremony” is an interpretation. “The experience was mystical” is another inference. The Menorca paper itself illustrates why those layers can diverge. ³¹

Fourth, model provenance quality independently from analytical strength. The Bes-vessel chemistry can be excellent while its exact excavation provenance is weak. Those dimensions should not be collapsed into one credibility slider. ³³

Fifth, track source drift. Mas Castellar is an excellent demonstration: “ergot fragments/microremains identified” can morph downstream into “hallucinogenic alkaloids chemically detected.” The interface should show the original wording beside the derivative claim. ⁴⁶

Sixth, attach alternative explanations to the same observation. Extraordinary Eleusinian testimony can support a pharmacological model, an intense-ritual model, or a mixed model. Aletheia should not allow a piece of evidence to appear exclusively beneath the hypothesis that first cited it. ⁴⁴

Seventh, preserve authorial confidence over time. The book's rhetoric, Muraresku's website, and his more qualified Harvard statements are not identical. Rather than deciding which is “really” his position, Aletheia can show the evolution. ⁴⁷

Eighth, display “resolution value” aggressively. The highest-value unresolved objects in this case are not another interpretive essay. They are: original Mas Castellar samples; securely contextualized Eleusinian vessels; Villa Vesuvio material; and securely identified early-Christian ritual cups. Direct modern analyses of those materials could change the state of the dispute dramatically. ⁴⁸

Ninth, keep modern advocacy separate from ancient evidence. Mosurinjohn and Ascough are right that modern psychedelic politics can influence the way ancient history is narrated; Muraresku himself connects historical precedent with contemporary possibilities. But whether either side has present-day religious or political motivations is a provenance/context field, not evidence that ergot was or was not in a fourth-century BCE cup. ⁴⁹

Finally, the headline verdict should remain deliberately conditional. The best summary for Aletheia today is:

Ancient ritual psychoactive use is no longer seriously in doubt as a general phenomenon. The ergot-kykeon hypothesis at Eleusis has acquired meaningful archaeological analogues and, in 2026, a credible experimental feasibility mechanism, making it a legitimate and testable historical hypothesis rather than a mere psychedelic fantasy. But direct evidence from Eleusis is still absent. The extension from Eleusis/Dionysian pharmacology to an early-Christian psychedelic Eucharist is substantially less evidenced: ancient capability, cultural overlap and literary parallels establish possibility, not transmission. The decisive next step is residue analysis of securely contextualized Eleusinian and early-Christian ritual vessels. 50

That, in my assessment, is where the disagreement actually lives—and it gives Aletheia something better than a pro/con bibliography: a set of claims whose uncertainty could genuinely be reduced by identifiable evidence.

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